

DATA 202 — Testing a Model · Week 6 Wednesday

Same penguins as today's slides: 342 penguins, split at random into 239 for training and 103 for testing. Your professor will tell you when to do each part.

1 · Five splits, five scores

Same model ($k = 5$), same penguins; only `random_state` changes.

split	random_state	test accuracy
ours	24	95.1%
another	15	92.2%
another	1	93.2%
another	7	97.1%
another	19	100.0%

- Lowest score: _____ %
Highest score: _____ %
- Circle** the split you'd be tempted to report.

2 · Build the confusion matrix

Our split, $k = 5$. The model got **46** Adelies, **18** Chinstraps and **34** Gentoos right. Its 5 mistakes:

bill length (mm)	bill depth (mm)	true species	predicted
44.4	17.3	Gentoo	Chinstrap
45.8	18.9	Adelie	Chinstrap
48.1	16.4	Chinstrap	Gentoo
40.9	16.6	Chinstrap	Adelie
42.9	17.6	Adelie	Chinstrap

- Fill in all nine cells. Rows = **true species**, columns = **predicted**.

	Adelie	Chinstrap	Gentoo
Adelie	_____	_____	_____
Chinstrap	_____	_____	_____
Gentoo	_____	_____	_____

- Accuracy = right $\div$ all = _____ $\div$ 103 = _____

3 · Finding the Chinstraps

1. In your matrix, **circle** the Chinstrap row and **box** the Chinstrap column.

2. **Recall** — of the real Chinstraps, how many did we find?

$$\text{recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

3. **Precision** — of the penguins we called Chinstrap, how many really are?

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

4. A lazy model answers *not Chinstrap* for every penguin. On “Chinstrap or not?”:

$$\text{accuracy} = \frac{\text{TN} + \text{FP}}{\text{TN} + \text{FP} + \text{TP} + \text{FN}} = \frac{\text{TN} + \text{FP}}{\text{TN} + \text{FP} + \text{TP} + \text{FN}} \cdot \text{recall} = \frac{\text{TN} + \text{FP}}{\text{TN} + \text{FP} + \text{TP} + \text{FN}}$$

5. In your matrix, write **FP** in each false-positive cell and **FN** in each false-negative cell for Chinstrap.